

one, of course, is that an IR LED emits infrared radiations which cannot be seen with the naked eyes, whereas a LED emits visible light. Another difference is that IR LED consumes a lot of current and thus they get damaged very fast. They do not have lives as long as LEDs.

An important precaution to be taken while using an IR LED is to not allow it to be forward biased for a very long time as we know that infrared radiations carry a lot of heat. Continuous IR radiations may heat the body or shell of the IR LED too much and can damage it also. Be careful not to touch IR LED after it has been forward biased for a long time since it may have become very hot.

2.3 Multiple voltage requirements

Almost all the electronic chips and devices which we will be using would work at different voltages. Multiple voltages requirement can be handled in several ways; the easiest and most straightforward of them is to use a different set of batteries for each main subsection. However, it is cumbersome and impractical to provide a different power source having different voltage ratings to each of the component. This method is useful when the motors used in the robot draw a lot of current. Motors generally distribute a lot of electrical noise throughout the power lines. And sometimes, this is the noise to which the electronic circuitry is extremely sensitive. Thus if you use different set of batteries for motors and circuitry, the electrical isolation provided by it to the circuitry almost completely eliminates the problems caused by noise.

In addition, when the motors are started initially, the excessive current drawn from the motors may deprive the electronic circuitry of the necessary current, which may induce 'sag' that can cause erratic circuit behaviour, which may in turn cause your robot's processor to lose control. Despite these advantages, using different batteries for different components makes the robot too heavy.

The other approach to handling multiple voltages is to use one main battery source and multiple voltage regulators to step it up or down according to voltage requirements of individual components in the system. This method is called DC-DC conversion. You can accomplish DC-DC conversion by using your own circuits or by purchasing specialty integrated circuit chips that make the job easier. A 9V battery's output voltage can be changed to a wide range of voltages – from as low as tending towards zero to as high as 15 V. Normally, a single battery will directly drive the motors,